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Latch system for microwave oven door

Abstract

A microwave oven includes a housing, a door, and a latch. The housing defines a first internal cavity configured to receive food. The door is rotatably secured to the housing, defines a second internal cavity, and has a handle. The door is configuring to pivot relative to the housing to transition the door between an open position and a closed position. The latch is disposed within the second internal cavity. The latch has a hook and a push button. The hook protrudes outward from second internal cavity and toward the housing. The hook is configured to engage and disengage the housing to lock and unlock the door to and from the housing, respectively. The push button is rigidly interconnected with the hook. The hook is configured to disengage the housing in response to depression of the push button.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to an appliance that is configured to cook food, such as a microwave oven.
BACKGROUND
(2) Microwave ovens may include doors that are rotatably secured to main portions of the microwave ovens.
SUMMARY
(3) A microwave oven includes a housing, a door, and a latch. The housing defines an internal cavity configured to receive food for cooking and an opening configured to provide access to the internal cavity. The door is rotatably secured to the housing, has external panels defining an internal pocket, and has a handle. The door is configuring to pivot relative to the housing in response to a user engaging the handle to transition the door between an open position and a closed position. The door provides access to the opening in the open position and covers the opening in the closed position. The latch is pivotably secured to the door within the internal pocket and configured to pivot about an axis relative to the door. The latch has a latch hook and a push button. The latch hook protrudes outward from the internal pocket along a first of the external panels and extends

toward the housing. The latch hook is configured to engage the housing to lock the door to the housing in the closed position and is configured to disengage the housing to unlock the door from the housing. The first of the external panels faces toward the housing. The push button protrudes outward from the handle and is rigidly interconnected to the latch hook within the internal pocket. The push button and the latch hook are configured to collectively pivot about the axis in response to depression of the push button to disengage the latch hook from the housing.

(4) A microwave oven includes a housing, a door, and a latch. The housing defines a first internal cavity configured to receive food for cooking. The door has a panel rotatably secured to the housing and a handle protruding outward from the panel. The door is configuring to pivot relative to the housing to transition the door between an open position and a closed position. The door provides access to the first internal cavity in the open position and covers the first internal cavity in the closed position. The panel and the handle collectively define a second internal cavity. The latch is disposed within the second internal cavity. The latch has a hook and a push button. The hook protrudes outward from the second internal cavity, through the panel, and toward the housing. The hook is configured to engage the housing to lock the door to the housing in the closed position and is configured to disengage the housing to unlock the door from the housing. The push button protrudes outward from the handle and is rigidly interconnected to the hook within the second internal cavity. The hook is configured to disengage the housing in response to depression of the push button.

(5) A microwave oven includes a housing, a door, and a latch. The housing defines a first internal cavity configured to receive food for cooking. The door is rotatably secured to the housing, defines a second internal cavity, and defines a pocket handle extending upward from a lower edge. The door is configuring to pivot relative to the housing to transition the door between an open position and a closed position. The door provides access to the first internal cavity in the open position and covers the first internal cavity in the closed position. The latch is disposed within the second internal cavity. The latch has a hook and a push button. The hook protrudes outward from the second internal cavity, beyond and external surface of the door, and toward the housing. The hook is configured to engage the housing to lock the door to the housing in the closed position and is configured to disengage the housing to unlock the door from the housing. The push button extends downward into the pocket handle and is rigidly interconnected to the hook within the second internal cavity. The hook is configured to disengage the housing in response to depression of the push button.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a front isometric view of a microwave oven;

(2) FIG. **2** is a front isometric view of a first embodiment of a door for the microwave oven;

(3) FIG. **3** is a rear isometric view of the first embodiment of the door for the microwave oven;

(4) FIG. **4** is a cross-sectional view taken along line **4-4** in FIG. **3**;

(5) FIG. **5** is a front isometric view of a first embodiment of a latch for the microwave oven door;

(6) FIG. **6** is a rear isometric view of the first embodiment of the latch for the microwave oven door;

(7) FIG. **7** is a lower rear isometric view of a second embodiment of the door for the microwave oven;

(8) FIG. **8** is a cross-sectional view taken along line **8-8** in FIG. **7**;

(9) FIG. **9** is a top isometric view of a second embodiment of a latch for the microwave oven door; and

(10) FIG. **10** is a bottom isometric view of the second embodiment of the latch for the microwave

oven door.

DETAILED DESCRIPTION

(11) Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments may take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the embodiments. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

(12) Referring to FIG. **1**, a front isometric view of a microwave oven **10** is illustrated. The microwave oven **10** includes a housing **12**. The housing includes a plurality of panels or walls **14** that define an internal cavity **16** in which food may be placed or received for cooking. The plurality of walls **14** may include a top wall, a bottom wall, and three side walls. The housing **12** may further define an opening **20** configured to provide access to the internal cavity **16**. The microwave oven **10** also includes a door **18** that is rotatably attached or secured to the to the housing **12**. For example, the door **18** may be rotatably secured or attached to the housing **12** via hinges **22** along a first end or first side **24** of the door **18**.

(13) A handle (not shown in FIG. **1**) may be secured to or defined along a second end or second side **26** of the door **18**. The door **18** may comprise a panel **28** and said handle, where the panel **28** may be rotatably secured or attached to the housing **12** via the hinges **22**. More specifically, the panel **28** may be comprised of a plurality of plates, external panels, or subpanels that are secured to each other and define an internal pocket or cavity (not shown in FIG. **1**). For example, the panel **28** may comprise a plurality of metal plates or sheet metal subpanels that are secured to each other and define an internal pocket or cavity. The door **18** is configuring to pivot relative to the housing **12** via the hinges **22** in response to a user engaging the handle to transition the door between an open position **30** and a closed position **32**. The door **18** provides access to the opening **20** and internal cavity **16** when in the open position **30**. The door **18** covers the opening **20** and internal cavity **16** when in the closed position **32**. The door **18** may be one of several embodiments described in further detail below. Therefore, the door **18** as illustrated in FIG. **1** should not be construed as limiting.

(14) Referring to FIG. **2-6**, a first embodiment of the door **18′** and a corresponding latching system for the door **18′** are illustrated. The door **18′** includes the panel **28** and a handle **34** protruding outward from the panel **28**. The door **18′** also includes a latch **36** that protrudes outward from the panel **28** along an opposing side of the panel **28** relative to the handle **34**. As previously stated, the panel **28** may more specifically be comprised of a plurality of plates, external panels, or subpanels **38** that are secured to each other and define an internal cavity or pocket **40**. The handle **34** may protrude outward from a first of the subpanels **38** that faces away from the housing **12** while the latch **36** may protrude outward from a second of the subpanels **38** that faces toward the housing **12**.

(15) The latch **36** is disposed within the internal pocket **40**. The latch **36** is pivotably secured to the door **18′** within the internal pocket **40** and is configured to pivot about an axis **42** relative to the door **18′**. More specifically, the latch **36** may include a pivot pin **44** about which the latch **36** pivots relative to the door **18′**. Each end of the pivot pin **44** may be disposed within a bushing hole or orifice **46** defined by the door **18′** internally within the internal pocket **40**. The pivot pin **44** may be rotatable about the axis **42** within the bushing holes or orifices **46**. The latch **36** includes a latch hook **48** and a push button **50**.

(16) The latch hook **48** protrudes outward from the internal pocket **40** along one of the external panels **38** and extends toward the housing **12**. The latch hook **48** also extends beyond an external surface of the door **18′**. More specifically, the latch hook **48** extends toward a front panel **52** of the housing **12**. The latch hook **48** is configured to engage the housing **12** to lock the door **18′** to the housing **12** in the closed position **32**. More specifically, the latch hook **48** is configured to extend through a notch, groove, or slot **54** defined by the front panel **52** of the housing **12** and a flat stopping surface **56** that extends upward along aback side of the latch hook **48** is configured to engage a backside of the front panel **52** to lock the door **18′** to the housing **12** in the closed position **32**. A spring **58** may bias the latch hook **48** upward so that the flat stopping surface **56** is aligned with the backside of the front panel **52** when the latch hook **48** is extended through the slot **54**, and so that the door **18′** automatically remains locked to the housing **12** when the door **18′** is in the closed position **32**. The latch **36** may include a protrusion **59** that functions to retain the spring **58** so that spring **58** remains in contact with the latch **36**. The spring **58** may engage both the latch **36** and an internal surface within the door **18′**, or more specifically an internal surface within the handle **34**.

(17) The latch hook **48** is configured to disengage the housing **12** to unlock the door **18′** from the housing **12**. More specifically, to unlock the door **18′** from the housing **12** while the door **18′** is in the closed position **32**, the latch hook **48** is rotated downward via the pivot pin **44** so that the flat stopping surface **56** is no longer aligned with the backside of the front panel **52** but is instead aligned with the slot **54** defined by the front panel **52** of the housing **12**. Once the flat stopping surface **56** is aligned with the slot **54** the door **18′** may freely transition between the open position **30** and the closed position **32**.

(18) The latch hook **48** may be rotated downward via the pivot pin **44** so that the flat stopping surface **56** is aligned with the slot **54** by depressing the push button **50**. The push button **50** may be rigidly interconnected to the latch hook **48** within the internal pocket **40**. The push button **50** and the latch hook **48** are configured to collectively pivot via the pivot pin **44** about the axis **42** in response to depression of the push button **50** to disengage the latch hook **48** from the housing **12** (e.g., to align flat stopping surface **56** with the slot **54**) in order to unlock the door **18′** from the housing **12**. The latch hook **48** may also include a ramped surface **60**. The ramped surface **60** may engage the front panel **52** of the housing **12** proximate to or within the slot **54** while the door **18′** is being transitioned to the closed position **32** so that the latch hook **48** is forced downward, allowing the latch hook **48** to extend through the slot **54** without the push button **50** being depressed. Once the latch hook **48** is beyond the slot **54**, the latch hook **48** is forced upward via the spring **58** to lock the door **18′** to the housing **12** in the closed position **32**. The ramping surface **60** allows the door **18′** to transition to the closed position **32** and automatically lock without depressing the push button **50**.

(19) Rigidly interconnected may refer to components that are connected or secured to each other such that relative positions of the components to each other remain the same, particularly under operating conditions. For example, if the position of a first component is changed, the position of a second components that is rigidly interconnected with the first component will also change so that the relative positions of the first and second components to each other remain the same. Multiple components formed as a single solid piece are rigidly interconnected. Multiple components secured to each other without moveable joints are rigidly interconnected. Relative positions of rigidly interconnected components may be considered to remain the same in the event minor deflections of the components occur when external forces are applied to the components, so long as such minor deflections do not affect the functionality of the components. Such minor deflections may include deflections that are not detectable or observable by a person.

(20) The push button **50** protrudes outward from the handle **34**. More specifically, the push button **50** extends outward from the handle **34** through a notch, groove, or slot defined by the handle **34**. The latch **36** includes a plurality of linking arms **62** rigidly interconnecting the latch hook **48** to the

push button **50**. The latch hook **48**, push button **50**, and plurality of linking arms **62** may form a single solid component. At least two of the plurality of linking arms **62** are non-linear relative to each other. A first and a second of the linking arms **62** may be substantially perpendicular to each other. Substantially perpendicular may refer to any incremental angle is that is between exactly perpendicular and 15° from exactly perpendicular.
(21) The handle **34** may be C-shaped. The handle **34** may have a central portion **64** that is spaced-apart from the panel **28**. More specifically, the central portion **64** may be spaced apart from the first of the subpanels **38** that faces away from the housing **12**. The handle **34** may also have opposing ends **66** that are each secured to the panel **28**. More specially, the opposing ends **66** may each be secured to the first of the subpanels **38** that faces away from the housing **12**.
(22) The handle **34** may define an internal channel **68** that is in communication within the in the internal pocket **40**. The internal channel **68** may also be said to form a portion of the internal pocket **40**. The push button **50** is rigidly interconnected to the latch hook **48** via the linking arms **62**, which extend through the internal channel **68** and/or the internal pocket **40**. The push button **50** more specifically may protrude outward from an external surface **70** of the central portion **64** of the handle **34**. The external surface **70** may be one any of the external surfaces of the handle **34**, but may more specifically be the external surface of the central portion **64** of the handle **34** that faces toward the panel **28**. More specially, the external surface **70** may face the first of the subpanels **38** that faces away from the housing **12**.
(23) Referring to FIG. **7-10**, a second embodiment of the door **18″** and a corresponding latching system for the door **18″** are illustrated. The door **18″** includes the panel **28** and defines a pocket handle **72** along a lower or bottom edge **74** of the door **18″**. The door **18″** also includes latch **76** that protrudes outward from the panel **28**. As previously stated, the panel **28** may more specifically be comprised of a plurality of plates, external panels, or subpanels **38** that are secured to each other and define an internal cavity or pocket **40**.
(24) The latch **76** is disposed within the internal pocket **40**. The latch **76** is pivotably secured to the door **18″** within the internal pocket **40** and is configured to pivot about an axis **78** relative to the door **18″**. More specifically, the latch **36** may include a pivot pin **80** about which the latch **76** pivots relative to the door **18″**. Each end of the pivot pin **80** may be disposed within a bushing hole or orifice **82** defined by the door **18″** internally within the internal pocket **40**. The pivot pin **80** may be rotatable about the axis **78** within the bushing holes or orifices **82**. The latch **76** includes a latch hook **84** and a push button **86**.
(25) The latch hook **84** protrudes outward from the internal pocket **40** along one of the external panels **38** and extends toward the housing **12**. The latch hook **84** also extends beyond an external surface of the door **18″**. More specifically, the latch hook **84** extends toward the front panel **52** of the housing **12**. The latch hook **84** is configured to engage the housing **12** to lock the door **18″** to the housing **12** in the closed position **32**. More specifically, the latch hook **84** is configured to extend through the notch, groove, or slot **54** defined by the front panel **52** of the housing **12** and a flat stopping surface **88** that extends upward along aback side of the latch hook **48** is configured to engage a backside of the front panel **52** to lock the door **18″** to the housing **12** in the closed position **32**. One or more springs **90** may bias the latch hook **84** upward so that the flat stopping surface **88** is aligned with the backside of the front panel **52** when the latch hook **84** is extended through the slot **54**, and so that the door **18″** automatically remains locked to the housing **12** when the door **18″** is in the closed position **32**. The latch **76** may include protrusions **92** that function to retain the springs **90** so that springs **90** remain in contact with the latch **76**. The springs **90** may engage both the latch **76** and an internal surface within the door **18″**, or more specifically an internal surface within the pocket handle **72**.
(26) The latch hook **84** is configured to disengage the housing **12** to unlock the door **18″** from the housing **12**. More specifically, to unlock the door **18″** from the housing **12** while the door **18″** is in the closed position **32**, the latch hook **84** is rotated downward via the pivot pin **80** so that the flat

stopping surface **88** is no longer aligned with the backside of the front panel **52** but is instead aligned with the slot **54** defined by the front panel **52** of the housing **12**. Once the flat stopping surface **88** is aligned with the slot **54** the door **18″** may freely transition between the open position **30** and the closed position **32**.

(27) The latch hook **84** may be rotated downward via the pivot pin **80** so that the flat stopping surface **88** is aligned with the slot **54** by depressing the push button **86**. The push button **86** may be rigidly interconnected to the latch hook **84** within the internal pocket **40**. The push button **86** and the latch hook **84** are configured to collectively pivot via the pivot pin **80** about the axis **78** in response to depression of the push button **86** to disengage the latch hook **84** from the housing **12** (e.g., to align flat stopping surface **88** with the slot **54**) in order to unlock the door **18″** from the housing **12**. Depression of the push button **86** may include pulling the push button **86** in a direction away from the housing **12**. The latch hook **84** may also include a ramped surface **94**. The ramped surface **94** may engage the front panel **52** of the housing **12** proximate to or within the slot **54** while the door **18″** is being transitioned to the closed position **32** so that the latch hook **84** is forced downward, allowing the latch hook **84** to extend through the slot **54** without the push button **86** being depressed. Once the latch hook **84** is beyond the slot **54**, the latch hook **84** is forced upward via the one or more springs **90** to lock the door **18″** to the housing **12** in the closed position **32**. The ramping surface **94** allows the door **18″** to transition to the closed position **32** and automatically lock without depressing the push button **86**.

(28) The push button **86** extends downward from the internal pocket **40** and into the pocket handle **72**. More specifically, the push button **86** extends downward into pocket handle **72** through a notch, groove, or slot defined by an internal surface within the pocket handle **72**. The latch **76** includes a plurality of linking arms **96** rigidly interconnecting the latch hook **84** to the push button **86**. The latch hook **84**, push button **86**, and plurality of linking arms **96** may form a single solid component. At least two of the plurality of linking arms **96** may be non-linear relative to each other.

(29) The microwave oven **10** may include a microwave generating device, such as a magnetron or a solid-state device. The microwave oven **10** may include a waveguide that defines a pathway or channel on an opposing side of a wall of the plurality of walls **14** relative to the internal cavity **16**. The wall of the plurality of walls **14** may define an orifice that establishes communication between the internal cavity **16** and the pathway or channel. A waveguide cover may be disposed over the orifice within the internal cavity **16**. The pathway or channel of the waveguide is configured to direct microwaves from the microwave generating device, through the waveguide cover, and to the internal cavity **16** in order to cook any food that is disposed within the internal cavity **16**.

(30) The microwave oven **10** may also include a power supply, such as a transformer, that provides electrical power to the microwave generating device, a capacitor, and a cooling fan. The cooling fan may be configured to cool the various components of the microwave oven **10**, such as the microwave generating device, power supply, capacitor, etc. Please note that for illustrative purposes, the electrical connections between the various components of the microwave oven **10** and the electrical connection between the microwave **10** and an external power source (e.g., an electrical plug and outlet connection) are not shown.

(31) The electronic components (e.g., microwave generating device, fan motors, power supply, capacitors, etc.) of the microwave oven **10** may be connected to a control panel, such as a human machine interface (HMI), and a controller, so that an operator may control various parameters. For example, the operator may be configured to input a cooking time, a cooking temperature, a desired mode of cooking (e.g., microwave cooking, defrost, etc.).

(32) The controller may be part of a larger control system and may be controlled by various other controllers throughout the microwave oven **10**. It should therefore be understood that the controller and one or more other controllers can collectively be referred to as a “controller” that controls various functions or components of the microwave oven **10** in response to signals from various sensors to control the various functions or components of the microwave oven **10**. The controller

may include a microprocessor or central processing unit (CPU) in communication with various types of computer readable storage devices or media. Computer readable storage devices or media may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the CPU is powered down. Computer-readable storage devices or media may be implemented using any of a number of known memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or any other electric, magnetic, optical, or combination memory devices capable of storing data, some of which represent executable instructions, used by the controller in controlling the microwave oven **10**.
(33) It should be understood that any component having a callout number that includes one or prime symbols (′) should be construed as having the same structure, subcomponents, and functionality as a component that includes the same callout number but without the one or prime symbols, unless otherwise stated or illustrated herein. It should also be understood that the designations of first, second, third, fourth, etc. for any component, state, or condition described herein may be rearranged in the claims so that they are in chronological order with respect to the claims. Furthermore, it should be understood that any component, state, or condition described herein that does not have a numerical designation may be given a designation of first, second, third, fourth, etc. in the claims if one or more of the specific component, state, or condition are claimed.
(34) The words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments may be combined to form further embodiments that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics may be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and may be desirable for particular applications.

Claims

1. A microwave oven comprising: a housing defining (i) an internal cavity configured to receive food for cooking and (ii) an opening configured to provide access to the internal cavity; a door (i) rotatably secured to the housing, (ii) having external panels defining an internal pocket, and (iii) having a handle, wherein the door (i) is configuring to pivot relative to the housing in response to a user engaging the handle to transition the door between an open position and a closed position, (ii) provides access to the opening in the open position, and (iii) covers the opening in the closed position; and a latch (i) pivotably secured to the door within the internal pocket and (ii) configured to pivot about an axis relative to the door, the latch having a latch book (i) protruding outward from the internal pocket along a first of the external panels, (ii) extending toward the housing. (iii) configured to engage the housing to lock the door to the housing in the closed position, and (iv) configured to disengage the housing to unlock the door from the housing, where the first of the external panels faces toward the housing, and push button (i) protruding outward from the handle and (ii) rigidly interconnected to the latch hook within the internal pocket to maintain relative positions between the push button and the lack hook, wherein the push button and the latch. hook are configured to collectively pivot about the axis while maintaining the relative positions in response to depression of the push button to disengage the latch hook from the housing.
2. The microwave oven of claim 1, wherein the latch includes a plurality of linking arms rigidly

interconnecting the latch hook to the push button.
3. The microwave oven of claim 2, wherein the latch hook, push button, and plurality of linking arms form a single solid component.
4. The microwave oven of claim 2, wherein at least two of the plurality of linking arms are non-linear relative to each other.
5. The microwave oven of claim 1, wherein the handle (**1**) is C-shaped, (ii) has a central portion that is spaced-apart from a second of the external panels such that a space is defined between the central portion and the second of the external panels and (in) bas opposing ends each secured to the second of the external panels, wherein the second of the external panels faces away from the housing.
6. The microwave oven of claim 5, wherein the handle definies an internal channel that is in communication with the internal pocket, and wherein the push button is rigidly interconnected with the latch hook via first and second linking arms extending through the internal channel and the internal pocket.
7. The microwave oven of claim 6, wherein the first and second linking arms are substantially perpendicular to each other.
8. The microwave oven of claim 5, wherein the push button protrudes outward from an external surface of the central portion of the handle and into the space.
9. The microwave oven of claim 8, wherein the external surface faces toward the space and the second of the external panels of the door.
10. The microwave oven of claim 1, wherein the handle is a pocket handle defined along a bottom edge of the door.
11. The microwave oven of claim 10, wherein the push button is disposed within the pocket handle.
12. A microwave oven comprising: a housing defining a first internal cavity configured to receive food for cooking; a door having (i) a panel rotatably secured to the housing and (ii) a handle protruding outward from the panel, wherein the door (i) is configuring to pivot relative to the housing to transition the door between an open position and a closed position, (ii) provides access to the first internal cavity in the open position, (ii) covers the first internal cavity in the closed position, and wherein the panel and the handle collectively define a second internal cavity; and a latch disposed within the second internal cavity, the latch having a hook (i) protruding outward from the second internal cavity, through the panel, and toward the housing, (ii) configured to engage the housing to lock the door to the housing in the closed position, and (iii) configured to disengage the housing to unlock the door from the housing, and a push button (i) protruding outward from the handle and (ii) rigidly interconnected to the hook within the second internal, cavity to maintain relative positions between the push button and the hook, wherein the hook is configured to disengage the housing in response to depression of the push button.
13. The microwave oven of claim 12, wherein the push button and the hook are configured to collectively pivot about an axis relative to the panel while maintaining the relative positions in response to depression of the push button to disengage the hook from the housing.
14. The microwave oven of claim 12, wherein the push button protrudes outward from an external surface of the handle, into a space defined between the handle and the panel, and toward the panel.
15. The microwave oven of claim 14, wherein the external surface faces toward the panel.
16. The microwave oven of claim 12, wherein the latch includes a plurality of linking arms (i) disposed within the second internal cavity and (ii) rigidly interconnecting the hook to the push button.
17. The microwave oven of claim 16, wherein the hook, push button, and plurality of linking arms form a single solid component.
18. A microwave oven comprising: a housing defining a first internal cavity configured to receive food for cooking; a door (i) rotatably secured to the housing, (ii) defining a second internal cavity, and (ii) defining a pocket handle extending upward from a lower edge, wherein the door (i) is

configuring to pivot relative to the housing to transition the door between an open position and a closed position, (ii) provides access to the first internal cavity in the open position, (iii) covers the first internal cavity ju the closed position; and a latch disposed within the second internal cavity, the latch having a hook (i) protruding outward from the second internal cavity, beyond an external surface of the door, and toward the housing, (ii) configured to engage the housing to lock the door to the housing in the closed position, and (iii) configured to disengage the housing to unlock the door from the housing, and a push button (i) extending downward into the pocket handle and (ii) rigidly interconnected to the hook within the second internal-cavity, cavity to maintain relative positions between the push button and the hook, wherein the hook is configured to disengage the housing in response to depression of the push button.
19. The microwave oven of claim 18, wherein the push button and the hook are configured to collectively pivot about an axis relative to the door while maintaining the relative positions in response to depression of the push button to disengage the latch hook from the housing.
20. The microwave oven of claim 19, wherein the latch includes a plurality of linking arms (i) disposed within the second internal cavity and (ii) rigidly interconnecting the hook to the push button.